IS1212 : Probability and Statistics

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Probability distribution of a discrete random variable

- The **probability distribution** for a random variable describes how probabilities are distributed over the values of the random variable.
- For a discrete random variable X with possible values $x_1, x_2, \ldots, x_n$ the probability mass function (or pmf) is,

$$\boldsymbol{f(x_i) = P(X = x_i)}$$

- Usually denoted by $\boldsymbol{f(x)}$.

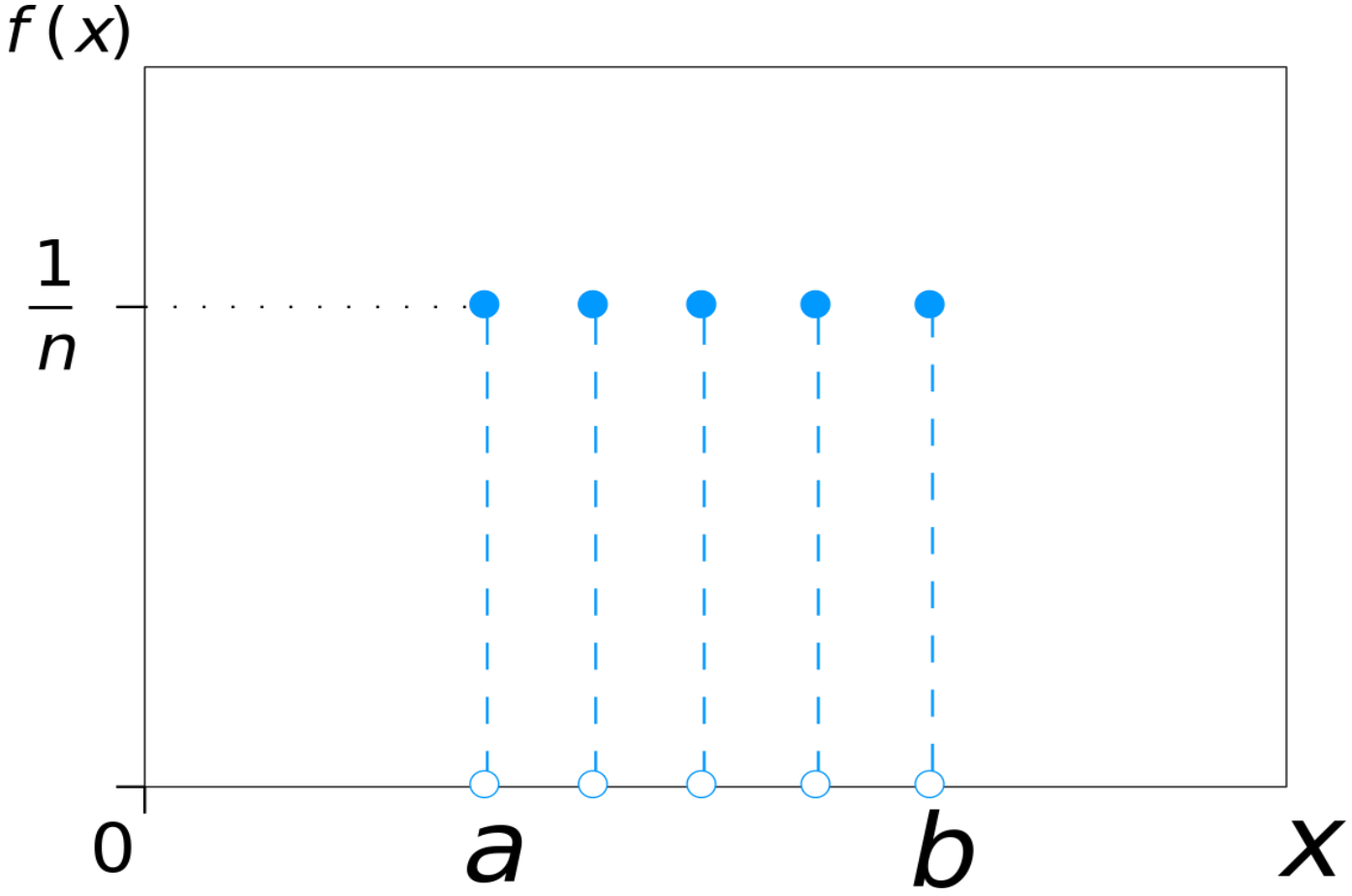
Discrete Probability Distribution

Conditions:

$$1)\ \ \mathbf{0} \leq \boldsymbol{P}(\boldsymbol{X} = \boldsymbol{x}) \leq \mathbf{1}$$

$$2)\ \ \sum_{all\ x} \boldsymbol{P}(\boldsymbol{X} = \boldsymbol{x}) = \mathbf{1}$$

Discrete Uniform Distribution



Discrete Uniform Probability Distribution

- This is the simplest discrete probability distribution.

- Here we assume that for n different possible values of a discrete random variable, each value has equal chance of occurrence.

- Therefore, the probability associated with each value of the random variable is,

$$\boldsymbol{P(X = x) = \frac{1}{n}}$$

Example

- Consider the experiment of rolling a fair die. We define the random variable X to be the number that appears on the upward face.

- For this experiment $n = 6$

- Thus, the probability distribution function for the discrete random variable X is,

$$f(x) = 1/6$$
$$for\ x = 1, 2, 3, 4, 5, 6$$

x	$f(x)$
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Example

The probability distribution for the random variable X is given below

x	$f(x)$
20	0.20
25	0.15
30	0.25
35	0.40

1. Is this probability distribution valid? Explain.
2. What is the probability that $X = 30$?
3. What is the probability that X is less than or equal to 25?
4. What is the probability that X is greater than 30?

Expected value of Uniform Distribution

- Suppose X is a random variable from discrete uniform distribution.

$$\boldsymbol{E(X) = (n + 1)/2}$$

Variance of uniform distribution

- Suppose X is a random variable from discrete uniform distribution

$$\boldsymbol{Var\,(x) \;=\; (n^2 - 1)\,/\,12}$$

Bernoulli Distribution

- Consider a random experiment,where there are only two mutually exclusive outcomes.
- One of them is called a “success” and the other is called a “failure”.
- Success is simply assigned to the outcome in which we are interested.

Example

- Consider a situation, when checking the quality of a product, the item selected at random is either non-defective or defective.
- If the study is on defective items, then the outcome "a defective item" is considered as a success.
- The outcome "a good item" is considered as a failure.

Bernoulli Trial

- In a trial, if,
  - it has only two possible outcomes
  - the probability of success is constant and
  - trials are independent when repeated

then that is called a Bernoulli trial!

Examples:

- Tossing a coin

  Success – Obtaining head

  Failure - Obtaining tail
- Rolling a die

  Success – Getting an even number

  Failure – Getting an odd number

Let p be the proportion of defective items in a lot.

Select an item at random from this lot and observe its quality.
This is usually known as a Bernoulli trial.
Then, $P\{a\ success\} = p$ and $P\{a\ failure\} = 1 - p$

The distribution of probabilities can be written as,

Number of success$[\boldsymbol{X = x}]$	**0**	**1**
Probability $P\{X = x\}$	$1 - p$	p

- This distribution is known as a ***Bernoulli distribution*** with parameter p.

Probability mass function of Bernoulli Distribution

$$P(X = x) = \begin{cases} p & ; if\ x = 1 \\ 1 - p & ; if\ x = 0 \\ 0 & ; o/w \end{cases}$$

$$P(X = x) = \begin{cases} p^{x}\,(1 - p)^{1-x} & ; x = 0,1 \\ 0 & ; o/w \end{cases}$$

Mean and Variance of Bernoulli Distribution

$$E[X] = 0\ x\ (1-p) + \ 1\ x\ p \ = \ p \ = \ probability\ of\ a\ success$$

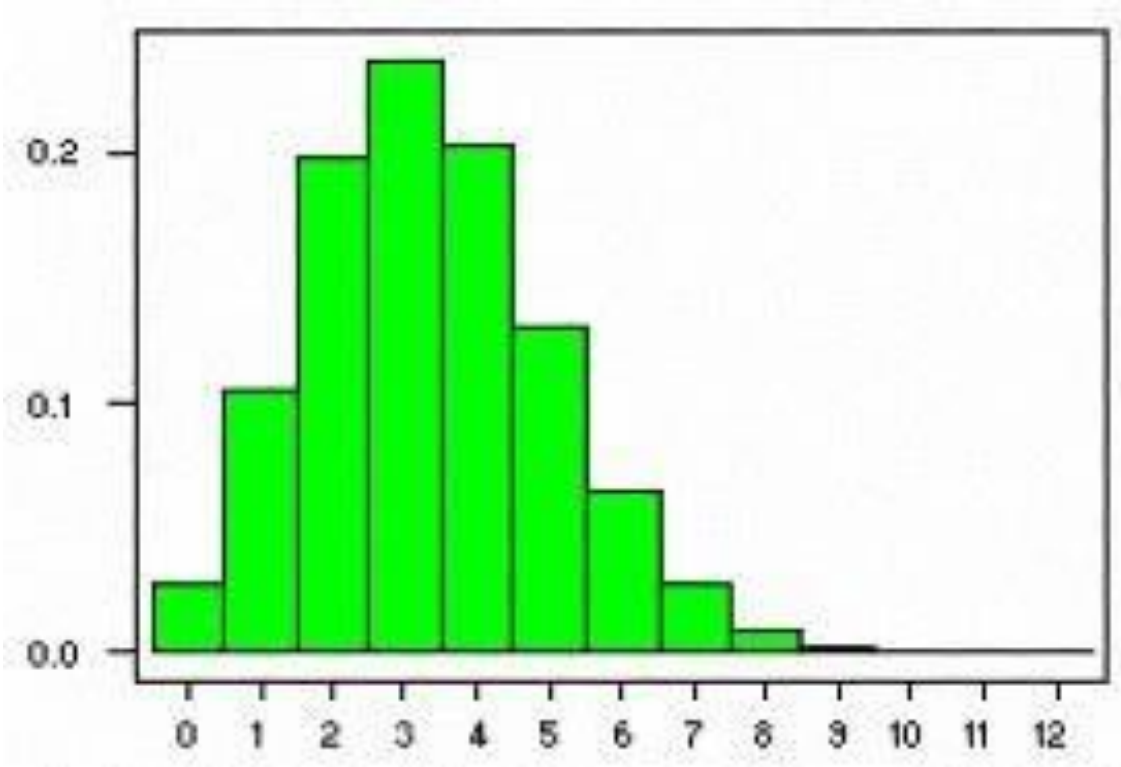
$$\boldsymbol{E[X] \ = \ p}$$

$$E[X^2] = \ 0^2\ x\ (1-p) + \ 1^2\ x\ p \ = \ p$$

$$Var\ (X) \ = \ E[X^2] - \ (E[X])^2 \ = p^2 - \ p \ = \ p\ (1-p)$$

$$\boldsymbol{Var\ (X) \ = \ p\ (1-p)}$$

Binomial Distribution



Binomial Experiments

- When Bernoulli trials are repeated ***n*** times, we say it forms a Binomial experiment.
- For a Binomial experiment,
  - each trial has only two possible outcomes
  - the probability of success is constant for all trials
  - trials are independent and
  - trials are repeated n times

Example: Tossing a coin 10 times

Binomial Probability Distribution

- Let's define the random variable X as the number of successes in a Binomial experiment.

- Then we say, X follows a

  Binomial Probability Distribution!

Parameters of a Binomial Distribution

- Number of times trials are repeated - n
- Probability of success - p
- Probability of failure → $q = 1-p$

$$X \sim Bin(n, p)$$

$$X = 0, 1, 2, \ldots, n$$

Probability mass function of Binomial Distribution

$$P(X = x) = {}^{n}C_{x}\ p^{x}\ (1 - p)^{(n - x)}$$

$$x = 0, 1, 2, \ldots, n$$

Example

The random variable X is distributed as Bin(4,0.25). Find the following probabilities correct to three decimal places.

i. $P(X = 3)$

ii. $P\ X > 2$

iii. $P(2 < X < 4)$

iv. $P(1 < X \leq 3)$

v. $P(X = 6)$

Expected Value and Variance for a Binomial Distribution

- If $X \sim Bin(n, p)$,

$$Mean = E(X) = np$$

$$\boxed{\boldsymbol{E(X) = np}}$$

$$Variance = V(X) = npq$$

$$\boxed{\boldsymbol{Var(X) = npq}}$$

$$Standrad\ deviation = \sqrt{npq}$$

Exercise 1

In a test, there are ten multiple choice questions. For each question, there is a choice of four answers and only one is correct. A student guesses each of the answers randomly.

a. Find the probability that only one answer is correct.

b. Find the probability that he gets 6 answers wrong.

c. He needs to obtain at least 7 answers correct to pass the exam. Find the probability that he will pass the exam.

Exercise 2

- The probability that it will be a fine day is 0.4.

a. Find the probability that 4 days will be fine in a given week.

b. Find the expected number of fine days in a week and the standard deviation.

Exercise 3

A university found that 20% of its students withdraw without completing the introductory English course. Assume that 20 students registered for the course.

a. Compute the probability that two or fewer will withdraw.

b. Compute the probability that exactly four will withdraw.

c. Compute the expected number of withdrawals.

Any questions?